

MULTI-USER AGENT ECOSYSTEMS • CLOSED-LOOP SKILL EVOLUTION

SkillClaw: Collective Skill Evolution for Multi-User LLM Agent Ecosystems

SkillClaw enables LLM agent skills to continuously evolve by aggregating cross-user interaction trajectories and applying agentic, evidence-driven skill updates.

Source: arxiv.org/abs/2604.08377

Figure: overview (figure_1.jpg)

PROBLEM

Agent Skills Remain Static After Deployment – Users Rediscover Solutions Independently

Static skill paradigm

- Skills are manually installed and rarely updated from runtime experience.
- Failures repeat across users; fixes do not accumulate at the ecosystem level.
- Knowledge stays local to a session instead of becoming shared infrastructure.

Evolving skill paradigm (SkillClaw)

- Collect interaction evidence across many users for the same skill.
- Diagnose root causes and propose verified mutations automatically.
- Synchronize improved skills back to all agents.

Core claim: static ecosystems prevent system-level knowledge accumulation

Closed-Loop Pipeline: From Multi-User Interaction to Shared Skill Evolution

MULTI-USER INTERACTION

Agents act → observe → recover

Structured session trajectories preserve action-feedback causal chains.

SHARED EVIDENCE BASE

Group trajectories by skill

Aggregate across users to expose behavioral boundaries and failure modes.

Evidence

STAGE 1

Analyze trajectories to identify patterns of success and failure.

Attribution

STAGE 2

Diagnose root causes; separate generalizable signal from idiosyncratic fixes.

Evolution

STAGE 3

Propose skill mutations: refine existing skills or create new ones.

Distribution

STAGE 4

Verify, merge, and synchronize updated skills back to all agents.

Reference figure: figure_1.jpg (paper overview)

Loop: evidence → attribution → evolution → distribution



Agentic Evolver: Evidence → Attribution → Evolution in Three Stages

1) Evidence analysis

Analyze skill-specific grouped trajectories to identify behavioral patterns (success paths + failure modes).

2) Attribution

Diagnose root causes and isolate generalizable improvements from one-off fixes.

3) Evolution

Propose mutations (refine an existing skill or create a new one) for verification and merge.

Not predefined rules

Open-ended agent reasoning over interaction evidence drives updates instead of hand-coded heuristics.

Key idea: autonomous agent reasoning over evidence

Output: verified skill mutations + sync



SIGNAL

Cross-User Trajectory Aggregation Exposes Skill Behavioral Boundaries

- Different users exercise the same skill under diverse contexts, producing complementary views of where it works and where it breaks.
- Aggregated trajectories preserve full action-feedback causal chains, enabling evidence-driven analysis.
- A single user rarely generates enough signal to distinguish a generalizable improvement from an idiosyncratic fix.
- SkillClaw is compatible with Claw-style systems: OpenClaw, CoPaw, IronClaw, PicoClaw, ZeroClaw, NanoClaw, NemoClaw.

Grouping: sessions → referenced skills → evidence base

Evaluation note: WildClawBench (Qwen3-Max backbone)

Multimodal Creative Pipeline Skill – 6 Days of Evolution Attempts

Conservative verification prevents regressions: only 1 of 6 candidates was accepted.

DAY	OUTCOME	RATIONALE (FROM LOG)
Day 1	Accepted	Validator accepted validate-tmp-workspace-inputs (inputs + environment setup) → upgraded best pool.
Day 2	Rejected	Validator rejected multimodal-input-validation-and-setup → kept current best pool.
Day 3	Rejected	Validator rejected multimodal-creative-task-pipeline candidate → kept current best pool.
Day 4	Rejected	Rejected improved pipeline (image classification + visual generation + structured validation) → kept current best pool.
Day 5	Rejected	Rejected pipeline + per-file fail-fast validation → kept current best pool.
Day 6	Rejected	Rejected extended PDF-to-poster / document-to-visual paths → not admitted to next cycle.

 CASE STUDY

Git Push Auth-Fallback Skill — Iterative Refinement Over 6 Days

Contrasts with creative pipeline: 3 of 6 days yielded accepted updates before plateau.



WRAP - UP

Key Takeaways: Three Properties That Distinguish SkillClaw from Existing Systems

